Implementing Identity and Access Management (IAM) Using Google Cloud

1. Project Title

Implementing Identity and Access Management (IAM) Using Google Cloud IAM

2. Objective

The objective of this project is to understand and implement Identity and Access Management (IAM) in a cloud environment. The project focuses on creating and managing users, assigning **role-based access control (RBAC)**, enforcing the **principle of least privilege**, and validating access restrictions using Google Cloud IAM.

3. Why IAM Is Important

Identity and Access Management (IAM) is a critical security mechanism in cloud computing that ensures **only authorized users** can access specific cloud resources. Improper access control can lead to:

- Data breaches
- Unauthorized modifications
- Insider threats

IAM helps organizations enforce security by defining **who can do what** on cloud resources.

4. Cloud Platform Used

- **Cloud Provider:** Google Cloud Platform (GCP)
- **Service Used:** Google Cloud IAM

- **Environment:** Google Cloud Skills Boost (Sandbox / Lab Environment)

A **sandbox environment** was used to safely implement and test IAM policies without any cost.

5. IAM Concepts Used

Concept	Description
Principal	An identity (user) that can access cloud resources
Role	A collection of permissions
Permission	An action allowed on a resource
RBAC	Role-Based Access Control
Least Privilege	Granting minimum permissions required

6. Project Architecture

- **User 1:** Project Owner (Administrator)
- **User 2:** Limited User (Viewer)
- **User 3:** Limited User (Editor)

Logical Steps:

1. Access Google Cloud Console using sandbox credentials.
2. Test multiple users with different roles.
3. Analyze IAM permissions for each role.
4. Validate access restrictions and RBAC policies.

7. Step-by-Step Implementation

Step 1: Accessing the Google Cloud Console

- Accessed Google Cloud Console using sandbox credentials.
- Opened **two separate browser tabs** for User 1 and User 2 (added User 3 in a third tab).

Step 2: Signing in as Project Owner (User 1)

- User 1 logged in with Owner credentials.
- Accessed **IAM & Admin Console**.
- Confirmed **Owner-level permissions**.

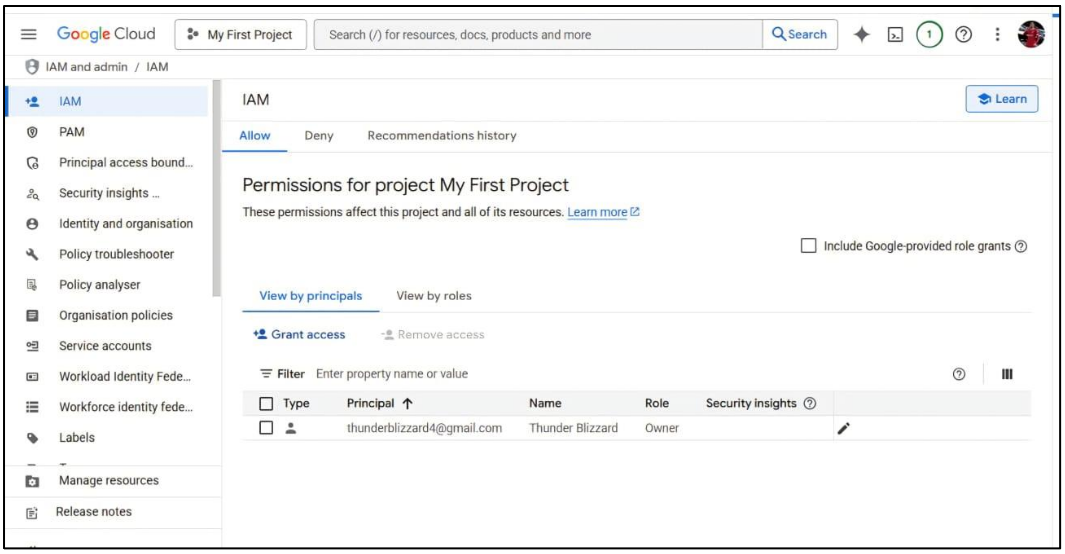


Fig. 1. IAM dashboard showing User 1 as Owner

Step 3: Exploring IAM Roles

Role	Permissions
Viewer	Read-only access
Editor	Modify resources but cannot manage IAM policies
Owner	Full administrative access including IAM management

- User 1 confirmed to have **Owner permissions**.

Fig. 2. IAM Roles List

Step 4: Signing in as Limited User (User 2 – Viewer)

- Logged in using User 2 credentials.
- Accessed IAM & Admin Console.
- Attempted to modify IAM policies and observed access restrictions.

Observation:

- Access **disabled**

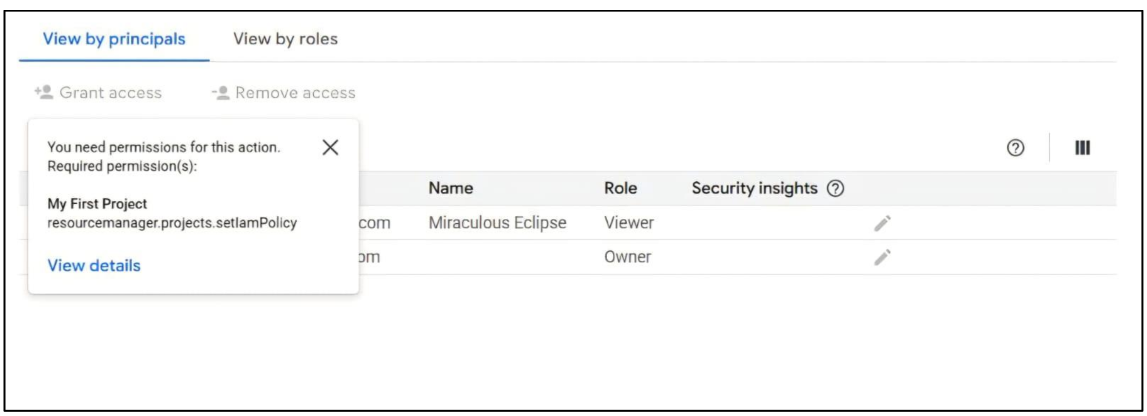


Fig. 3. Viewer role page, permission denied message

Step 5: Signing in as Limited User (User 3 – Editor)

- Logged in using User 3 credentials.
- Assigned **Editor role**.

- Tested access:
    - Able to **modify resources** (create/update instances, edit projects).
    - **Cannot manage IAM policies** (cannot add/remove users or roles).

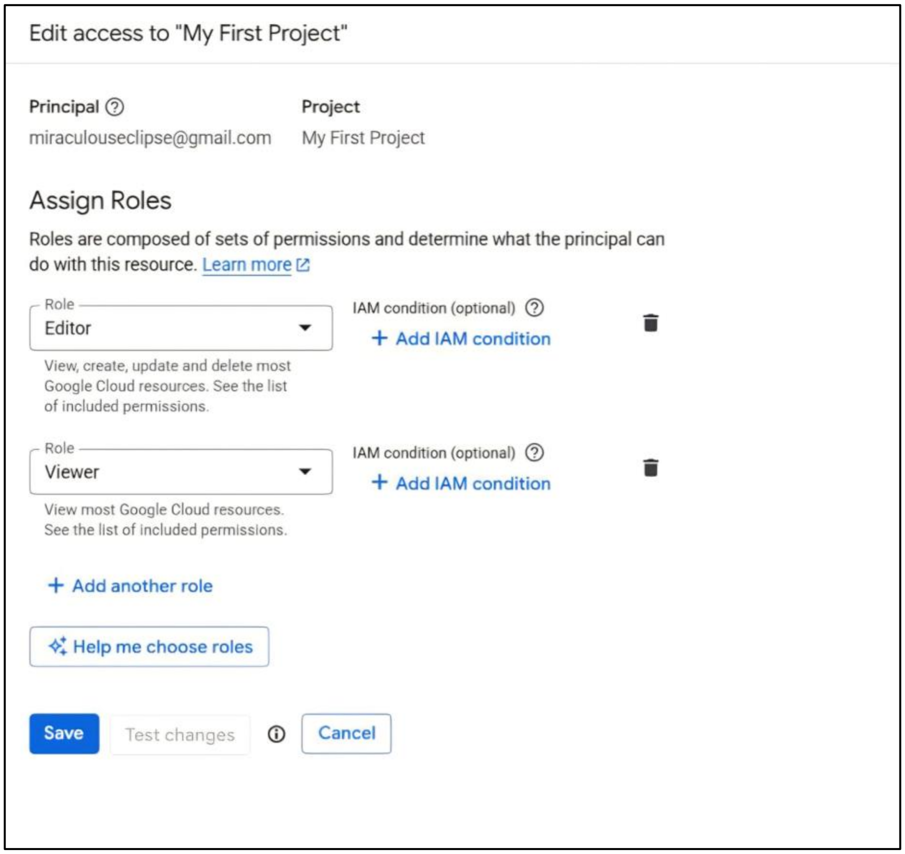


Fig. 4. Setting multiple roles for the same user

Step 6: Testing Role-Based Access Control (RBAC)

- **User 2 (Viewer):** Read-only access → cannot change resources
- **User 3 (Editor):** Modify access → can edit resources but cannot manage IAM
- **User 1 (Owner):** Full access → can manage IAM and assign roles

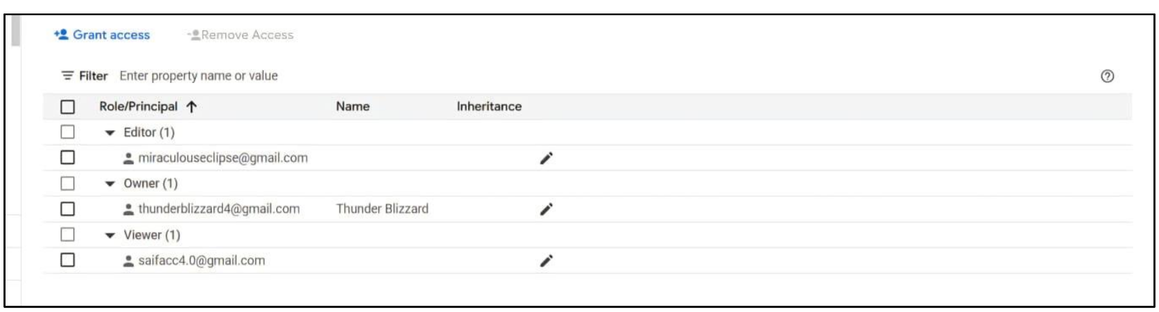
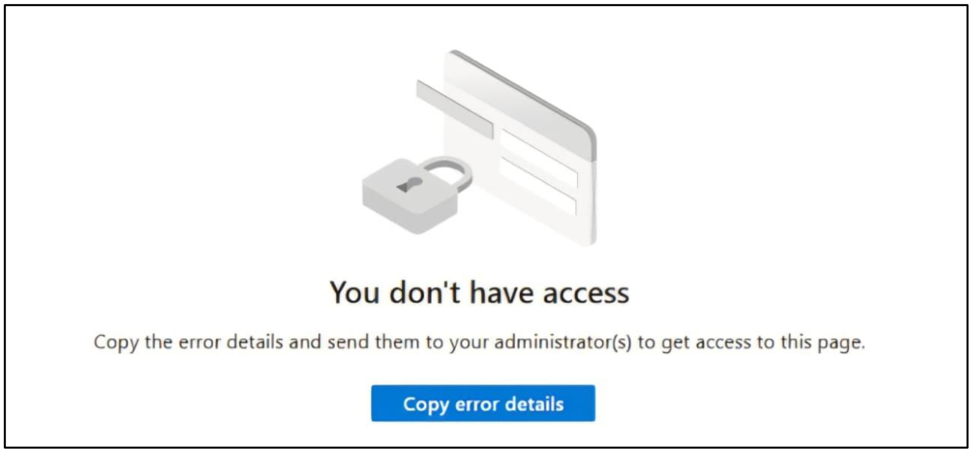


Fig. 5. Separation of duties

8. Security Validation & Observations

- Viewer users **cannot modify IAM policies**
- Editors **can modify resources** but not IAM policies
- Only Owners can **grant/revoke access**
- IAM enforces **permissions at the project level**
- Role misconfiguration can lead to **security risks**
- **Least privilege** significantly reduces attack surface



9. Principle of Least Privilege

- Viewer role assigned to User 2 → ensures **read-only access**
- Editor role assigned to User 3 → allows **resource editing** without full admin access
- Owner role remains with User 1 → full control for administrative tasks

This aligns with **industry security best practices**.

10. Result

- IAM system successfully enforced **role-based access control**
- Unauthorized users prevented from performing restricted actions
- Editor role verified as **intermediate permission level**
- Demonstrated the effectiveness of **least-privilege assignment**

11. Conclusion

- Successfully implemented **Identity and Access Management using Google Cloud IAM**
- Simulated multiple users with **different roles**
- Validated **RBAC and least-privilege enforcement**
- Confirmed IAM protects cloud environments from **unauthorized access and risks**

12. Skills Gained

- Cloud Identity & Access Management
- Role-Based Access Control (RBAC)
- Least Privilege Enforcement
- Cloud Security Fundamentals
- Access Control Testing